

Task 1

① $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$

$$\dim(\mathbb{R}^2) = 2, \quad \dim(\mathbb{R}^3) = 3$$
$$2 \neq 3$$

T is not invertible

② $T(a_1, a_2, a_3) = (3a_1 - 2a_3, a_2, 3a_1 + 4a_2)$

$$T(1, 0, 0) = (3, 0, 3)$$

$$T(0, 1, 0) = \cancel{(0, 1, 4)} (0, 1, 4)$$

$$T(0, 0, 1) = \cancel{(0, 0, 0)} (-2, 0, 0)$$

$$R(T) = \text{span} \{ (3, 0, 3), (0, 1, 4), (-2, 0, 0) \}$$

$$\text{rank}(T) = 3$$

$$\text{rank}(T) + \text{nullity}(T) = 3$$

$$\text{nullity}(T) = 0$$

T is one-to-one and onto

T is invertible

③

$$\dim(M_{2 \times 2}) = 4 \neq \dim(P_2) = 3$$

T is not invertible

Task 2

① $\dim(\mathbb{R}^3) = 3 \neq \dim(P_3) = 4$

$\mathbb{R}^3$ and $P_3(\mathbb{R})$ are not isomorphic

② $\dim(\mathbb{R}^4) = 4 = \dim(P_3(\mathbb{R})) = 4$

$\mathbb{R}^4$ and $P_3(\mathbb{R})$ are isomorphic

③ $V = \{A \in M_{2 \times 2}(\mathbb{R}) \text{ where } \text{tr}(A) = 0\}$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad a, b, c, d \in \mathbb{R}, \quad a+d=0$$

$$= \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

$$\text{span}(V) = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

$$\dim(V) = 3 = \dim(\mathbb{R}^3)$$

V and $\mathbb{R}^3$ is isomorphic

Task 3

① $A \in M_{n \times n}$

If A is invertible, there ~~exis~~ is a $B = A^{-1}$, $AB = BA = I$

$$A^2 = 0$$

$$BA^2 = B \cdot 0$$

$$BA \cdot A = 0$$

$$I \cdot A = 0$$

$$A = 0$$

$$A \cdot B = 0 \cdot B = 0 \neq I$$

$\therefore A$ is not invertible

② If A is invertible, exsist A^{-1} that $A \cdot A^{-1} = A^{-1} \cdot A = I$

$$AB = 0, \quad A^{-1}AB = A^{-1} \cdot 0 = IB = A^{-1} \cdot 0 = 0$$

$$IB = 0, \quad B = 0$$

B is nonzero matrix.

$\therefore A$ is not invertible

Task 4

β is linear independent

$T(\beta)$ is also linear independent

If $T(\beta)$ is not a basis for W

~~$\text{span}(W) =$~~

$W = \text{span}\{T(\beta) + S\}$, S and $T(\beta)$ is linear independent

$\therefore V = \text{span}\{\beta\}$

$$\dim(V) < \dim(W)$$

$\therefore T$ is not isomorphism if $T(\beta)$ is not a basis for W

If $T(\beta)$ is a basis for W

$$\dim(V) = \dim(W)$$

T is isomorphism if and only if $T(\beta)$ is a basis for W